

Create Rubric

40 points

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Q1 A1

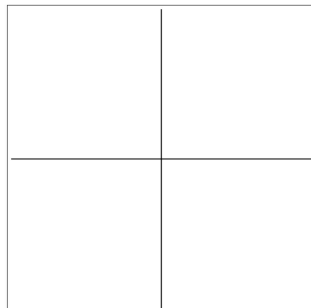
4 points

Q1.1 (a)

1 point

(a) Use the axes to the right to sketch the graph of one function $f(x)$ that satisfies all of the following conditions.

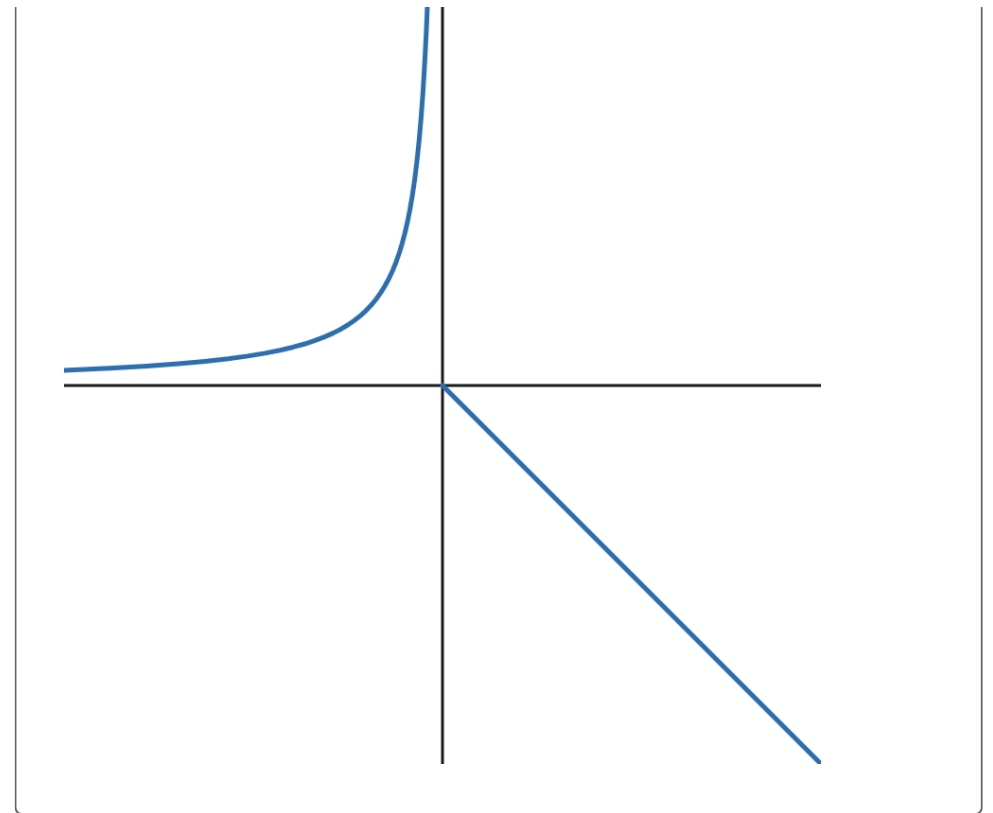
$$\begin{array}{ll} \lim_{x \rightarrow -\infty} f(x) = 0 & \lim_{x \rightarrow 0^+} f(x) = \infty \\ \lim_{x \rightarrow 0^+} f(x) = 0 & \lim_{x \rightarrow \infty} f(x) = -\infty \end{array}$$



1

-0.0

Correct. One possible answer is shown below.



2

-0.5

2 or 3 of the limits are satisfied by the graph, but not all 4.



3

-1.0

Less than 2 of the limits are satisfied by the graph.

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Q1.2 (b)(i)

1 point

(i) $\lim_{x \rightarrow 0^+} \sqrt{x} \left[1 + \sin^2 \left(\frac{2\pi}{x} \right) \right]$
Hint: $0 \leq \sin^2 \left(\frac{2\pi}{x} \right) \leq 1$



1

-0.0

Correct. The answer uses the Squeeze Theorem to show that the limit is 0.



2

-0.5

Some meaningful progress is made, but there are still significant mistakes.



3

-1.0

No meaningful progress is made.

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Q1.3 (b)(ii)

1 point

(ii) $\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$



1

-0.0

Correct. The answer shows the left and right and limits are -2 and 2 , respectively, so the limit does not exist.

2 -0.5

Some meaningful progress is made, but there are still significant mistakes.

3 -1.0

No meaningful progress is made.

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Q1.4 (c)

1 point

(c) For what value of the constant c is the function $g(x)$ continuous at $x = 2$?

$$g(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

1 -0.0

Correct. The answer is $c = \frac{2}{3}$.

2 -0.5

Some meaningful progress is made, but there are still significant mistakes.

⋮ 3 -1.0

No meaningful progress is made.

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Q2 A2

4 points

Q2.1 (a)(i)

1 point

(a) The graph of a function $f(x)$ is shown to the right.

(i) List five different integers x for which $f'(x) = 0$.

⋮ 1 -0.0

Correct

⋮ 2 -0.5

Some meaningful progress is made toward a correct answer, but there are still significant mistakes. The correct answer is $x = -4, -3, -2, 0, 2$.

Q2.2 (a)(ii)

0.5 points

- (ii) List two different integers x for which $f'(x)$ is negative.



3

-1.0

Little to no progress is made toward a correct answer.
The correct answer is $x = -4, -3, -2, 0, 2$.

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1

-0.0

Correct



2

-0.5

Incorrect. The correct answer is $x = -1, 3$.

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Q2.3 (b)

2.5 points

(b) Use the limit definition of the derivative to determine if the piecewise function

$$f(x) = \begin{cases} -2x - 9 & x < -3 \\ x^2 + 4x + 4 & x \geq -3 \end{cases}$$

is differentiable at $x = -3$. For full credit, you must use the limit definition of the derivative. Clearly show all of your work.

⋮ 1 **+2.5**
Correct

⋮ 2 **+0.5**
Correct left-hand limit is setup:
$$\lim_{h \rightarrow 0^-} \frac{f(-3 + h) - f(-3)}{h}.$$

⋮ 3 **+0.5**
Left-hand limit is evaluated to ∞ .

⋮ 4 **+0.5**
Correct right-hand limit is setup:
$$\lim_{h \rightarrow 0^+} \frac{f(-3 + h) - f(-3)}{h}.$$

⋮ 5 **+0.5**
Right-hand limit is evaluated to -2 .

⋮ 6 **+0.5**
Correctly concludes that $f'(-3)$ does not exist.



7

+0.0

No credit

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Q3 A3

4 points

Q3.1 (a)

1.5 points

(a) $f(x) = x^{\ln(x)}$



1

+1.5

Correct. The answer is $f'(x) = x^{\ln(x)} \left(\frac{2 \ln(x)}{x} \right)$.



2

+0.5

The natural logarithm is used to transform the right hand side into $\ln(x)^2$.



3

+0.5

The derivative of $\ln(x)^2$ is taken appropriately.

⋮ 4 +0.5

The derivative of $\ln(f(x))$ is taken appropriately, and both sides are multiplied by $f(x)$ (or if the natural base is used to transform the function, then the change rule is correctly used so that the factor $e^{\ln(x)^2}$ appears in the derivative).

⋮ 5 +0.0

No credit

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Q3.2 (b)

2.5 points

(b) $g(x) = \frac{e^x(x^2 + 1)}{(\tan x + \cos x)^3}$

⋮ 1 +2.5

Correct. The answer is

$$\frac{(\tan x + \cos x)^3(e^x(x^2 + 1) + 2xe^x) - 3e^x(x^2 + 1)(\tan x + \cos x)^2}{(\tan x + \cos x)^6}$$

⋮ 2 +0.5

The general framework of the quotient rule is used correctly (even if the derivatives are not taken correctly).

⋮ 3 +0.5

Some meaningful progress is made toward the derivative of the numerator, but there are still significant mistakes.

⋮ 4 +1.0

The derivative of the numerator is found correctly using the product rule.

⋮ 5 +0.5

Some meaningful progress is made toward the derivative of the denominator, but there are still significant mistakes.

⋮ 6 +1.0

The derivative of the denominator is found correctly using the chain rule.

⋮ 7 +0.0

No credit

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Q4 A4

4 points

Q4.1 (a)

2 points

- (a) Find a point on the curve $2y^5e^x = 3y - 6x$ at which the tangent line to the curve has a slope equal to 2, or show that no such point exists.



1

-0.0

Correct



2

-0.5

Some meaningful progress is made toward implicit differentiation of the equation, but there are still significant mistakes. The correct answer is

$$2y^5e^x + 10y^4e^x \cdot \frac{dy}{dx} = 3\frac{dy}{dx} - 6.$$



3

-1.0

Little to no progress is made toward implicit differentiation of the equation. The correct answer is

$$2y^5 e^x + 10y^4 e^x \cdot \frac{dy}{dx} = 3 \frac{dy}{dx} - 6.$$

⋮ -0.5

Some meaningful progress is made toward solving the equation for $\frac{dy}{dx} = 2$, but there are still significant mistakes. The correct answer is $(0, 0)$.

⋮ -1.0

Little to no progress is made toward solving the equation for $\frac{dy}{dx} = 2$. The correct answer is $(0, 0)$.

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Q4.2 (b)(i)

1 point

(b) Let $f(x) = \sqrt[3]{28 - x^2}$.

(i) Find the linear approximation of $f(x)$ at $x = 1$.

⋮ -0.0

Correct

⋮ -0.5

Some meaningful progress is made toward a correct answer, but there are still significant mistakes. The correct answer is $-\frac{2}{27}(x-1)+3$.

3 -1.0

Little to no progress is made toward a correct answer. The correct answer is $-\frac{2}{27}(x-1)+3$.

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Q4.3 (b)(ii)

1 point

(ii) Use your answer from part (i) to approximate $\sqrt[3]{27.19}$. You do not need to simplify your answer.

1 -0.0

Correct

2 -0.5

Some meaningful progress is made toward a correct answer, but there are still significant mistakes. The correct answer is $3 + \frac{1}{135}$.



3

-1.0

Little to no progress is made toward a correct answer.

The correct answer is $3 + \frac{1}{135}$.

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Q5 A5

4 points

Q5.1 (a)

1.5 points

- (a) The radius of a sphere is increasing at a rate of 4 mm/s. How fast is the volume increasing when the diameter is 80 mm? You don't need to simplify your answer, but you should include units. The volume of a sphere with radius r is

$$V = \frac{4}{3}\pi r^3$$



1

+1.5

Correct. The answer is $25600\pi \frac{\text{mm}^3}{\text{s}}$.



2

+0.5

The correct time derivative is taken:

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}.$$



3

+0.5

The correct values are substituted to obtain the numerical answer 25600π .



4

+0.5

The correct units are reported: $\frac{\text{mm}^3}{\text{s}}$.



5

+0.0

Little to no progress is made toward a solution.

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Q5.2 (b)

2.5 points

(b) A plane flies over an observer in such a way that the horizontal distance from the observer to the point on the ground directly below the plane is increasing at a rate of 400 miles per hour, and the altitude (height) of the plane is increasing at a rate of 30 miles per hour. Find the rate at which the angle of elevation from the observer to the airplane is changing when the plane is 3 miles above the ground and the straight line distance from the observer to the plane is $\sqrt{10}$ miles. You don't need to simplify your answer, but you should include units.



1

+2.5

Correct. The answer is -117 radians per hour.



2

+0.5

The left hand side of the equation $\tan \theta = \frac{y}{x}$ is differentiated correctly: $\sec^2 \theta \cdot \frac{d\theta}{dt}$.

⋮ 3 **+0.5**

The right hand side of the equation $\tan \theta = \frac{y}{x}$ is differentiated correctly: $\frac{x \cdot \frac{dy}{dt} - y \frac{dx}{dt}}{x^2}$.

⋮ 4 **+0.5**

The correct values for x and $\sec^2 \theta$ or $\cos^2 \theta$ are found ($x = 1$ and either $\sec^2 \theta = 10$ or $\cos^2 \theta = \frac{1}{10}$).

⋮ 5 **+0.5**

The correct numerical value of -117 is found after substituting all known information into the differentiated equation.

⋮ 6 **+0.5**

The correct units are reported: radians per hour.

⋮ 7 **+0.0**

Little to no progress is made toward a solution.

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Q6 B1

4 points

Q6.1 (a)

1.5 points

(a) What does the first derivative test say about the critical points $x = 0$ and $x = 5$?

⋮ 1 +1.5

Correct. The correct answer is that 0 is not an extrema and 5 is a local minimum.

⋮ 2 +0.5

The correct first derivative sign chart is constructed

⋮ 3 +0.5

0 is correctly identified as not an extrema (with correct reasoning). Credit is awarded if this answer correctly follows from an incorrect sign chart.

⋮ 4 +0.5

5 is correctly identified as a local minimum. Credit is awarded if this answer correctly follows from an incorrect sign chart.

⋮ 5 +0.0

Little to no progress is made toward a correct answer.

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Q6.2 (b)

1.5 points

(b) What does the second derivative test say about the critical points $x = 0$ and $x = 5$?

⋮ 1 +1.5

Correct. The correct answer is that the second derivative test is inconclusive for $x = 0$ and shows that $x = 5$ is a local minimum.

⋮ 2 +0.5

The second derivative is evaluated at $x = 0$ and $x = 5$.

⋮ 3 +0.5

It is correctly stated that the test is inconclusive for $x = 0$

⋮ 4 +0.5

It is correctly stated that the test shows that $x = 5$ is a local minimum.

⋮ 5 +0.0

Little to no progress is made toward a solution.

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Q6.3 (c)

1 point

(c) Find the absolute minimum and absolute maximum of $f(x)$ on the interval $[0, 4]$. Report your answers as ordered pairs.

⋮ 1 +1.0

Correct. The answer is $(0, 0)$ is the absolute minimum and $(3, 108)$ is the absolute maximum. Full credit is not awarded if more than $x = 0$, $x = 3$, and $x = 4$ are tested, or if $x = 4$ is not tested.

⋮ 2 +0.5

The correct points are tested: $x = 0$, $x = 3$, and $x = 4$.

⋮ 3 +0.5

Some unnecessary points are tested in the correct function, $f(x)$ (not $f'(x)$ or $f''(x)$), or not all necessary points are tested.

⋮ 4 +0.0

Little to no progress is made toward a correct answer.

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Q7 B2

4 points

Q7.1 (a)

2 points

Consider the function $f(x) = 1 - x^2 + \frac{1}{100}e^{2x}$.

- (a) Show that there are at least two points where the slope of $f(x)$ is equal to the value of $f(x)$, i.e., where $f'(x) = f(x)$.

⋮ 1 +2.0

Correct. The solution states $g(x) = f'(x) - f(x)$ is continuous, so we can apply the IVT. Then the solution

finds two disjoint intervals on which the function must have a root.

⋮ 2 +0.5

It is stated that $f'(x) - f(x)$ is continuous, so we may use the IVT.

⋮ 3 +1.0

Two changes in sign of $f'(x) - f(x)$ are detected by finding 3 appropriate values.

⋮ 4 +0.5

One change in sign of $f'(x) - f(x)$ is detected by finding 2 appropriate values.

⋮ 5 +0.5

Given the values used to detect the change in sign, the intervals in which the root lies are correctly determined.

⋮ 6 +0.0

No meaningful progress is made.

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Q7.2 (b)

2 points

(b) Show that there are at most two points where the slope of $f(x)$ is equal to the value of $f(x)$.



1

+2.0

Correct. The solution states $g(x)$ is differentiable, so we can apply the MVT (or Rolle's theorem). Then the solution shows that $g''(x)$ has no roots, so $g(x)$ has at most two roots.



2

+0.5

It is stated that $g(x)$ is differentiable, so we may use the MVT (or Rolle's Theorem).



3

+0.5

$g''(x)$ is correctly calculated.



4

+0.5

It is correctly reasoned that $g''(x) = 0$ has no solutions (e.g. via $g''(x)$ always being positive).



5

+0.5

It is correctly reasoned from $g''(x)$ having no roots that $g(x)$ has at most two roots.

⋮ 6 +0.0

No meaningful progress is made.

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Q8 B3

4 points

Q8.1 (a)

1 point

(a) What are the x -values of the local maxima/minima of f (if any)? Classify each as a maximum or minimum.

⋮ 1 -0.0

Correct. $x = -\sqrt{3}$ is a local minimum and $x = \sqrt{3}$ is a local maximum.

⋮ 2 -0.5

Some meaningful progress is made toward a solution, but there are still significant errors.

Q8.2 (b)

1 point

(b) What are the x -values of the points of inflection of f (if any)?



3

-1.0

No meaningful progress is made toward a solution.

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1

-0.0

Correct. $x = \pm\sqrt{6}$ are points of inflection.



2

-0.5

Some meaningful progress is made toward a solution, but there are still significant errors.



3

-1.0

No meaningful progress is made toward a solution.

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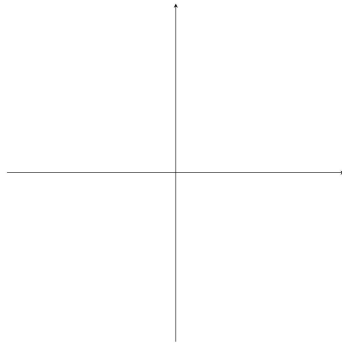
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Q8.3 (c)

2 points

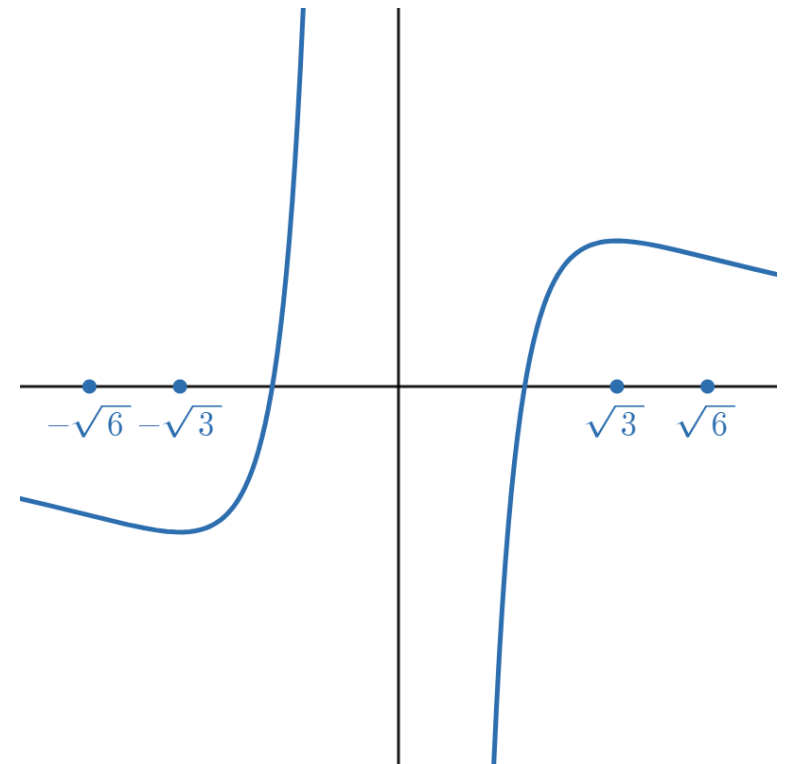
(c) Suppose we know that $f(x)$ has a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, and x -intercepts at $x = 1$ and $x = -1$. Sketch a graph of f below and label the points on the x -axis corresponding to any relative extrema and points of inflection.



1

+2.0

Correct. The answer is shown below.



2

+0.5

The asymptotes are graphed correctly.



3

+0.5

The x -intercepts are graphed correctly.



4

+0.5

The critical points are graphed and labeled correctly.



5

+0.5

The inflection points are graphed and labeled correctly.



6

+0.0

No meaningful progress is made.

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Q9 B4

4 points

Q9.1 (a)

1 point

(a) What is the minimum vertical distance between the parabolas $y = -8x^2 + 9x - 3$ and $y = -4x^2 + 17x + 6$.



1

+1.0

Correct



2

+0.5

The correct distance function is found:

$$D(x) = 4x^2 + 8x + 9.$$



3

+0.5

The distance function is correctly optimized: $(-1, 5)$.



4

+0.0

No meaningful progress is made toward an answer.

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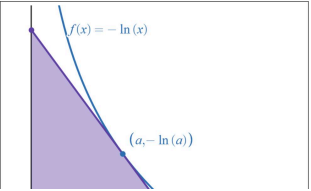
Q9.2 (b)

3 points

(b) A right triangle is formed from the x and y axes along with a line tangent to $f(x) = -\ln(x)$ at a point $x = a$ as shown in the picture. Use the following steps to find the value of a that maximizes the area of this triangle.

Step 1: Find an expression for the line tangent to $f(x)$ at some point a . Since a is unspecified, the expression for this line will depend on a .

Step 2: Find expressions for the x and y intercepts of the line from Step 1. These intercepts will also

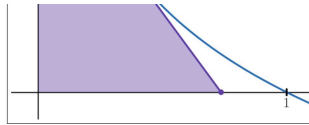


1

+3.0

be in terms of a .

Step 3: The intercepts from Step 2 are the width and height of the triangle, so use them to express the area in terms of a . Now use calculus to find the value of a that maximizes this area (a should be between 0 and 1 because $f(x)$ has an x intercept of 1 as shown in the picture).



Correct

⋮ +0.5

The equation of the tangent line is found:

$$y = -\frac{1}{a}(x - a) - \ln(a).$$

⋮ +0.5

The correct y -intercept is found: $1 - \ln(a)$

⋮ +0.5

The correct x -intercept is found: $a - a \ln(a)$.

⋮ +0.5

The correct formula for the area is found:

$$\frac{1}{2}a - a \ln(a) + \frac{1}{2}a \ln(a)^2.$$

⋮ +0.5

The derivative is found correctly: $-\frac{1}{2} + \frac{1}{2} \ln(a)^2$.

⋮ +0.5

The critical point e^{-1} is found correctly.

8 +0.0

No meaningful progress is made toward an answer.

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Q10 B5

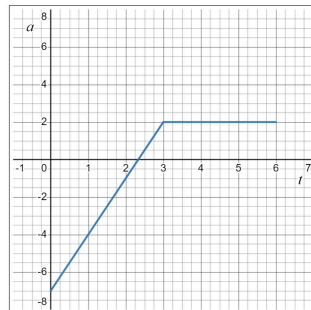
4 points

Q10.1 (a)

1 point

An object's acceleration (in m/s^2) at t seconds is given by the function $a(t)$ which is graphed to the right. Suppose the object's velocity at $t = 1$ is $3 m/s$.

(a) Find the object's velocity as a function of time, $v(t)$, from $t = 0$ to $t = 3$.



1 +1.0

Correct. The answer is $v(t) = 1.5t^2 - 7t + 8.5$.

2 +0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.

3 +0.0

****No meaningful progress is made toward a solution.

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Q10.2 (b)

1 point

(b) Find the object's change in position from $t = 0$ to $t = 3$. You don't need to simplify your answer.

⋮ 1 +1.0

Correct. The answer is 7.5.

⋮ 2 +0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.

⋮ 3 +0.0

No meaningful progress is made toward a solution.

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Q10.3 (c)

1 point



(c) Find the object's velocity as a function of time, $v(t)$, from $t = 3$ to $t = 6$.



1

+1.0

Correct. The answer is $v(t) = 2t - 5$.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.



3

+0.0

No meaningful progress is made toward a solution.

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Q10.4 (d)

1 point



(d) Find the object's position as a function of time, $s(t)$, from $t = 3$ to $t = 6$ if $s(3) = 0$.



1

+1.0

Correct. The answer is $s(t) = t^2 - 5t + 6$.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.



3

+0.0

No meaningful progress is made toward a solution.

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